

GARCH as a State-Space Model for Volatility

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1 Introduction: Why GARCH Exists

Financial returns appear deceptively simple:

$$r_t = \mu + \varepsilon_t$$

But the residuals ε_t are **not** i.i.d.

1.1 Empirical Stylized Facts

1. **Volatility clustering**: quiet periods follow quiet periods; wild periods follow wild periods
2. **Leverage effect**: negative shocks increase future volatility more than positive ones
3. **Heavy tails**: return distributions have excess kurtosis
4. **Time-varying correlations**: cross-asset dependencies change over time

1.2 The Core Insight

The real object of interest is not returns, but **conditional variance**:

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim \text{i.i.d.}(0, 1)$$

GARCH models specify **how** σ_t^2 **evolves**.

State-Space Interpretation:

- Returns are the **observation equation**
 - Volatility is a **latent state**
 - GARCH is a **nonlinear state-space model**
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2 Univariate GARCH Models

2.1 Symmetric GARCH(1,1): The Baseline

The canonical model:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Mechanism:

- Yesterday's shock ε_{t-1}^2 updates today's variance
- Yesterday's variance σ_{t-1}^2 persists

Key Properties:

Parameter	Interpretation
α	News sensitivity (shock impact)
β	Memory (variance persistence)
$\alpha + \beta$	Total persistence

Stationarity condition: $\alpha + \beta < 1$

Long-run (unconditional) variance:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}$$

Note: GARCH beats moving averages because it **responds asymmetrically in time**, not just magnitude.

2.2 Asymmetric Models: Capturing Leverage Effects

Empirical fact: Bad news moves volatility more than good news.

2.2.1 A-GARCH (Asymmetric GARCH)

Adds signed shocks via a shift parameter:

$$\sigma_t^2 = \omega + \alpha(\varepsilon_{t-1} - \gamma)^2 + \beta\sigma_{t-1}^2$$

2.2.2 GJR-GARCH

Adds an indicator for negative shocks:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \lambda\mathbf{1}_{\{\varepsilon_{t-1} < 0\}}\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

Comparison:

Model	Advantage
A-GARCH	Numerically easier to estimate
GJR-GARCH	More interpretable asymmetry parameter

Both capture the same economic phenomenon with different parameterizations.

2.3 E-GARCH: Log-Volatility as State

A conceptual leap—model **log variance** instead of variance:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha|z_{t-1}| + \gamma z_{t-1}$$

Advantages:

1. No positivity constraints needed
2. Asymmetry enters naturally
3. Volatility responds proportionally, not absolutely
4. Better behaved numerically
5. Often achieves best in-sample fit

Important: E-GARCH is particularly attractive for **option pricing** because it aligns with how implied volatility behaves.

2.4 Heavy Tails: Student-t GARCH

Even with GARCH, returns are too fat-tailed for Gaussian shocks.

Solution: Replace Gaussian innovations with Student-t:

$$z_t \sim t_\nu$$

What changes:

- Likelihood function
- One extra parameter: degrees of freedom ν

What does NOT change:

- Volatility dynamics
- Forecast equations

Division of Labor:

- **GARCH** handles time-dependence in volatility
- **Student-t** handles cross-sectional tail risk

Warning: Best in-sample fit $\neq$ best forecast performance!

2.5 Regime-Switching: Normal Mixture & Markov-Switching GARCH

Markets switch between regimes. Model this explicitly:

$$\varepsilon_t \mid \mathcal{F}_{t-1} \sim \sum_{k=1}^K \pi_k \mathcal{N}(0, \sigma_{k,t}^2)$$

Each regime has its own GARCH dynamics.

Interpretation:

Regime	Characteristics
“Normal”	Lower volatility, slower mean reversion
“Crisis”	Higher volatility, faster mean reversion

Note: This framework anticipates Hidden Markov Models (HMMs) and regime-switching state-space models.

3 Multivariate GARCH Models

3.1 The Curse of Dimensionality

For n assets, the covariance matrix has $\frac{n(n+1)}{2}$ unique elements.

Problem: Direct estimation is computationally infeasible for large n .

Solution: Impose **structure** on the covariance dynamics.

3.2 Factor GARCH (F-GARCH)

Assume returns are driven by a few common factors:

$$r_t = Bf_t + u_t$$

Approach: Model GARCH on factors, not individual assets.

Advantages:

- Captures systematic risk efficiently
- Dimensionality reduction
- Useful for large equity universes

Limitation: Idiosyncratic risk u_t must be modeled separately.

3.3 DCC-GARCH: Dynamic Conditional Correlations

Decompose the covariance matrix:

$$V_t = D_t C_t D_t$$

Where:

- D_t : diagonal matrix of GARCH volatilities
- C_t : time-varying correlation matrix

Key insight: Volatility and correlation are **separate states** with their own dynamics.

DCC updates correlations using standardized returns:

$$Q_t = (1 - a - b)\bar{Q} + a(z_{t-1}z'_{t-1}) + bQ_{t-1}$$

$$C_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}$$

Note: DCC-GARCH is extremely practical for FX and commodity portfolios.

3.4 Orthogonal GARCH (O-GARCH)

Use PCA to rotate the system:

$$X_t = WP_t$$

Approach: Model GARCH on **principal components**, not original assets.

Why this works:

1. Principal components are uncorrelated by construction
2. Only a few PCs explain most variance
3. Computationally efficient

Natural Applications:

- Term structure modeling
 - Energy futures curves
 - Yield curve dynamics
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4 Practical Applications

4.1 Simulation

Generate realistic Monte Carlo paths with:

- Volatility clustering
- Fat tails
- Regime dependence

4.2 Value-at-Risk (VaR)

Volatility-adjusted approaches:

- **Historical simulation:** Scale historical returns by current σ_t
- **Monte Carlo VaR:** Simulate paths using GARCH dynamics

4.3 Portfolio Optimization

Replace static covariance with GARCH-based time-varying covariance:

$$\Sigma_t = \mathbb{E}[r_t r_t' | \mathcal{F}_{t-1}]$$

4.4 Hedging

Warning: Empirical evidence shows:

- Limited improvement over static hedges
- High turnover costs
- Best suited when state-dependence is economically significant

5 Summary

5.1 Key Takeaway

GARCH is **not** just “a volatility model.”

It is a **latent-state system** with:

1. **Feedback** — shocks affect future volatility
2. **Memory** — volatility is persistent
3. **Partial observability** — we observe returns, not volatility

Use GARCH when **state-dependence matters**, not as a plug-in variance estimate.

5.2 Model Selection Guide

Model	Use Case
GARCH(1,1)	Baseline, forecasting
GJR/A-GARCH	Equity indices (leverage effect)
E-GARCH	Option pricing, log-volatility applications
Student-t GARCH	Heavy-tailed assets
DCC-GARCH	Multi-asset portfolios
O-GARCH	High-dimensional systems with factor structure